FLEET TEAM DIGEST

737MAX-FTD-33-20001

Issue Title : Burnt/Melted Service Light Lenses at the Main Wheel Well

Airplane Model	ATA	Minor Model(s)
737MAX	3332-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
03/05/2020	08/05/2022	06/30/2022

Estimated Completion Date	Next Update Date

Revision Description

Updated milestones and final action.

Applicability

All 737MAX airplanes

Description

The service and cargo lighting system provides illumination in various service compartments for ground maintenance and inspection purposes, and for cargo loading/unloading operations. Boeing started receiving reports of damaged lenses on the LED Service Light assembly P/N 81000-65401 in the main wheel well. The lenses appear burnt and melted, opening a hole in the middle of the lens. Please see the attachments for examples of this condition. The main wheel well contains four P/N 81000-65401 service lights: L172, L173, L174 and L175. While this light is also installed in other unpressurized locations in the airplane, no reports of melted or discolored lenses have been reported for service lights installed in those areas.

Status

Boeing has completed the root cause investigation on the burned lenses. It was determined that the root cause of the burned lens reported in service was due to the light's self-generated heat during normal operations and not from an external heat source or from a chemical interaction with corrosive contaminations. The polycarbonate lens material has a relatively low softening temperature. Engineering lab testing had demonstrated that this temperature was exceeded and caused the lens to melt when the lens was partially blocked or fully blocked to simulate the dirty contaminated lens condition as seen in service. The exterior surface of the lens was covered with dirt and debris trapping the heat energy radiated from the LEDs and caused the part of the lens directly on top of the LEDs to melt due to increased heat absorption over time.

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A design solution has been defined and validated with a successful engineering test. The design solution involved replacing the existing flat lens with a curved lens to create some distance between the lens and the light source to eliminate the heating effect from the LEDs. Production implementation and availability of the new lens P/N 81111-65401 to support fleet repair was completed in June 2022.

In addition, Koito revised the service light CMM 33-30-01 to add a lens removal/replacement instruction along with making the current lens available for spares purchases for customers who wish to replace the damaged lens on their own. The revision was released on 26-May-2020.

Interim Action

Boeing recommends airline operators to clean the lenses of the main wheel well service light assembly periodically to remove the accumulation of debris and dust to mitigate the lens melting issue.

Boeing also requests operators airlines to check their maintenance record and/or perform a quick inspection of their 737MAX fleet to find similar lens melting issues that may have gone unnoticed before.

Final Action

Production cut in of the light assembly with the revised lens was completed in June 2022.

Koito Service Bulletin 81000-33-060 which provides instructions to convert the existing service light P/N 81101-65401 (with flat lens) to an improved service light P/N 81101-65401, Mod. A (with curved lens) is available.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Established: February 2020 (Complete)

Solution Selected: May 2020 (Complete)

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CMM Update: May 2020 (Complete)

Production Incorporation: June 2022 (Complete)

Parts Available: June 2022 (Complete)

Supplier Service Bulletin: August 2022 (Complete)

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Service Light Assembly	S283A306-1	Koito	81000-65401

Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability

Attachments:

a) Lens Melting Example 2

b) Lens Melting Example

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